

CAN THE CAMERA TELL? ACTION-SET IDENTIFIABILITY AND PRINCIPLED AB- STENTION FOR PHYSICAL 3D GEOMETRY

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ABSTRACT

A corridor seen through a doorway, the same corridor reflected in a mirror, and a photograph of it shown on a screen can produce the same image. The three differ in what a hand would touch, so a depth predictor answering from one viewpoint is guessing between them rather than measuring. This paper asks a prior question: given the camera motions an observer can actually perform, is the physical explanation recoverable at all? The separability of two explanations is defined as how differently they predict the scene will look after a given motion, and a scene is called identifiable when some available motion separates every explanation still in play. When none does, the system reports that instead of answering. A benchmark is built in which competing explanations are pixel-identical at the starting view, so chance-level single-view accuracy becomes a measured property of the data rather than an assumption, and in which some scenes are unresolvable by design, making correct abstention something that can be scored. On real photographs a calibrated stereo pair supplies one known camera motion, and a check is introduced that discards images whose reference depth cannot settle the question being asked. Across published depth models spanning several architectures, the proposed measures are reported alongside deliberately trivial baselines, and a small model reading a frozen encoder’s features is shown to predict when that encoder will be wrong, including on architectures it was never trained on.

1 INTRODUCTION

A corridor viewed through an open doorway, the same corridor reflected in a planar mirror, and a photograph of it displayed on a screen can be arranged to produce an identical image. The three scenes differ in the surface a hand would touch, which is empty air in the first case, a pane of glass in the second and a rigid panel in the third, yet at a single viewpoint the light reaching the sensor is the same. The physical explanation of the observed geometry is therefore not determined by the image, and any answer produced from that image alone is selected by a prior rather than by evidence.

This distinction carries practical weight wherever perception is coupled to action. A service robot must separate a doorway from its reflection before a path is committed, an assistive system is expected to report the surface in front of a user rather than the scene depicted on it, and an augmented-reality device places content on a panel rather than inside the room that panel shows. In each case the cost of an error is set by physical contact, so the quantity of interest is the geometry of the first real surface along the ray, and the failure of interest is a confident report of geometry that is not present.

From accurate depth to identifiable depth. Monocular depth estimation now recovers relative geometry reliably across diverse scenes, and large pre-trained predictors transfer to unfamiliar imagery with little adaptation. Several complementary lines of work extend perception to exactly the configurations described above. Multi-layer and order-aware formulations represent more than one surface along a ray, so that a transparent interface and the scene behind it are both retained. Dedicated detectors recover mirrors and glass as first-class scene elements. Active and next-best-view perception treats observer motion as a resource to be allocated toward informative viewpoints.

Uncertainty estimation supplies a scalar that accompanies each prediction, selective prediction converts such a scalar into a decision to answer or to defer, and split conformal prediction supplies finite-sample guarantees on how often deferral occurs.

Each of these establishes a capability on which the present work depends, and together they address what the depth is, which surfaces are present, where the camera should move next, and how certain a prediction is. The matched configurations above raise a further question, which is the subject of this paper: is the physical explanation of this visual geometry recoverable at all from the observations that this observer is able to make?

Why this is a distinct question. Accuracy is a property of a prediction and uncertainty is a property of a predictor, whereas identifiability is a property of the triple formed by the competing physical explanations, the actions available to the observer, and the resolution at which differences become perceptible. A view-tracked display observed within its aperture is geometrically indistinguishable from the scene it depicts, and a static planar mirror reflecting a static room produces content parallax matching that of a real room seen through an opening of the same shape. In both cases the answer is not merely difficult to obtain, but absent from the available evidence. Enlarging a model or collecting further images from the same viewpoint leaves the situation unchanged, whereas a suitable change of viewpoint resolves it immediately. Identifiability is defined here relative to an action set for this reason, and the label “not identifiable” is treated as a statement about that triple rather than as a further physical hypothesis. The framing yields a prediction that can be tested independently of architecture: where the evidence cannot decide, a scalar confidence has no mechanism by which to register the absence of evidence, because the same image remains consistent with every hypothesis.

Contributions.

1. **Action-set identifiability as a decision rule.** Separability between two image-formation hypotheses is defined as the distance between the consequences they predict for an intervention, and identifiability as the best such separability attainable over an available action set. Falling below a perceptual threshold licenses abstention, which turns recoverability into an explicit and computable precondition for answering.
2. **A matched-counterfactual benchmark whose premise is measured rather than assumed.** Competing mechanisms are pixel-identical at the reference view to a maximum deviation of order 10^{-14} px, so chance-level single-view performance is a verified property of the construction. Cases that no permitted action resolves are included by design, which makes correct abstention a scorable outcome.
3. **A weakest-pair criterion for selecting the intervention.** Actions are chosen to maximise the smallest separability among hypothesis pairs still in contention, rather than a belief-weighted sum over all pairs. The criterion is T -optimality from optimal design for model discrimination (Atkinson & Fedorov, 1975; Hunter & Reiner, 1965), applied here to active visual perception and compared against the summed objective on this problem class.
4. **A protocol for real photographs, and the ground-truth admissibility criterion it requires.** Calibrated stereo is treated as a one-element action set, predictions are calibrated outside the ambiguous region and scored inside it, and images whose reference measurement cannot arbitrate the question are identified and excluded.
5. **Evidence across model families, and a decodable signal.** Behaviour is characterised over published checkpoints spanning several architecture families, with trivial baselines reported alongside the proposed measures, and a policy fitted on features pooled from a frozen encoder transfers to held-out architectures.

2 RELATED WORK

Depth on mirrors, glass and displays. Mirrors are recovered as first-class scene elements with dedicated instance masks and plane annotations, enabling depth to be refined on surfaces where standard reconstruction fails (Tan et al., 2021), and transparency is modelled by decomposing radiance into interface, transmitted and reflected components (Na et al., 2026). Benchmarks quantify the difficulty directly, covering specular and transparent surfaces with dense annotation (Ramirez et al.,

2023), large-scale transparent objects (Chen et al., 2022), and simulation-to-real depth restoration for such materials (Dai et al., 2022). These establish that the surfaces studied here are recognised as difficult and that recovering their geometry is an active target. The present work takes a step back from recovering the geometry and asks first whether the mechanism producing an appearance is recoverable from the observations available.

Multi-layer and order-aware geometry. Representing more than one surface along a ray is well established. Ordinal multi-layer benchmarks supply relations for both the first surface and the scene behind it, together with real and synthetic training data (Wen et al., 2025); self-determined grouping formulates transparency as unordered events along a ray (Wen & Deng, 2026); and steerable predictors condition on a focus distance to select which layer is returned (Kim et al., 2026). This line supplies the representational capacity that the present formulation assumes, and one such benchmark is used here as an independent evaluation set whose annotations share no machinery with the measures defined in Section 3.

Apparent versus physical geometry. That predictors report depicted rather than physical structure is documented at scale by a benchmark spanning illusion categories including printed pictures, screens and mirrors (Tang et al., 2025). The distinction between apparent and physical geometry is therefore taken as established. What is added here is a criterion determining when that distinction is decidable from the observations an observer can obtain, together with an explicit abstention when it is not, and a benchmark constructed so that such undecidable cases are present and scorable.

Active perception and experimental design. Selecting viewpoints to maximise expected information is long established in active and next-best-view perception, where the objective is typically a belief-weighted sum over competing possibilities. The question posed here is complementary: rather than which view is most informative, it asks whether any view within a bounded action set is informative enough to license a commitment. The criterion used for that purpose is not new. Designing experiments to discriminate between rival models is a mature area of statistics, in which the maximin criterion of equation 8 is known as T -optimality (Atkinson & Fedorov, 1975; Hunter & Reiner, 1965); its role here is the transfer of that criterion into active visual perception.

Causal identifiability, selective and conformal prediction. Identifiability under temporal interventions is established in causal representation learning (Lippe et al., 2022); that interventional framing is borrowed and applied to the narrower question of which optical mechanism produced an image, with the action set playing the role of the intervention family. Deferring on uncertain inputs follows Chow’s rule (Chow, 1970), and split conformal prediction supplies distribution-free, finite-sample coverage guarantees (Vovk et al., 2005; Lei et al., 2018). Those guarantees constrain how many predictions are retained rather than which ones, so a calibrated procedure inherits the ranking of the score it wraps. The abstention rule used here is derived instead from a geometric condition on the available evidence, and conformal prediction is retained as a baseline against which that condition is compared.

3 METHOD

The method maintains a set of competing physical explanations for an observed image, predicts what each implies after a candidate camera motion, measures how far those predictions diverge, and answers only when some available motion is guaranteed to separate every explanation still in contention.

3.1 HYPOTHESES AND THE ACTION SET

Let I_0 be an image captured at a reference pose C_0 , and $F_0 = E(I_0)$ the geometry a perception encoder E reports for it, represented as landmarks carrying image coordinates, depths and visibility flags. Instead of returning F_0 alone, a family of image-formation hypotheses is maintained: H_D asserts the structure is physically present, H_E that it is depicted on a flat emissive panel, H_R that it is a virtual image in a planar mirror, and H_T that it is seen through a planar transparent slab. These four are chosen because they occur in ordinary indoor environments and each places the first touchable surface in a different location, which is what a contact-based cost function distinguishes.

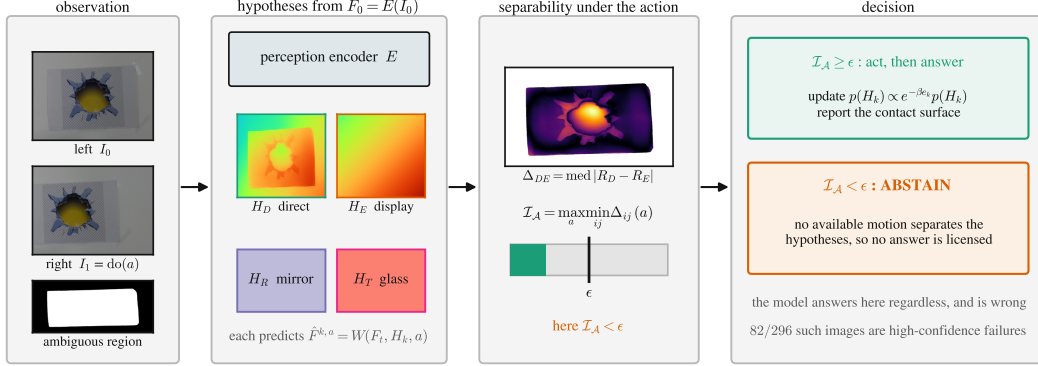


Figure 1: **The method on one real image.** A stereo pair supplies an executed action, and the annotated region marks where the surface is ambiguous. The encoder’s geometry gives H_D directly and, fitted to a plane over the same region, H_E ; the mirror and glass hypotheses enter the same way where the scene admits them. Warping the second view under each hypothesis gives the separability Δ_{DE} of equation 13, and identifiability equation 5 is compared against the perceptual threshold. Here it falls short, so equation 6 declines to answer, while the predictor answers and is wrong. Depth maps are from a published checkpoint used as released.

The hypotheses are matched by construction, meaning they produce pixel-identical observations at C_0 . Matching isolates the effect of interest: were they to differ in appearance at the reference view, a single-view predictor could separate them from that difference, and any later gain from moving the camera could not be attributed to the motion.

The observer acts. An action $a \in SE(3)$ is a rigid camera motion applied as an explicit intervention,

$$\text{do}(C_1 = C_0 \cdot a), \quad (1)$$

where C_0 is the reference pose, C_1 the resulting pose, and $\cdot$ denotes composition of rigid transforms. The $\text{do}(\cdot)$ notation replaces conditioning because the viewpoint is chosen rather than encountered, so the evidence obtained is attributable to the action. Actions come from a bounded, enumerable set $\mathcal{A}$ of translations and small rotations under a fixed motion budget, giving $|\mathcal{A}| = 33$ here; the budget is modest because the setting is a device making a small corrective movement rather than one repositioning freely. Finiteness is what allows identifiability to be evaluated by enumeration before any motion is executed, which is what makes abstention possible in advance of acting.

3.2 PREDICTING THE CONSEQUENCE OF AN ACTION

Each hypothesis maps current geometry and a candidate action to a predicted observation,

$$\hat{F}^{k,a} = W(F_t, H_k, a), \quad (2)$$

where F_t is the geometry estimated at time t , H_k the k -th hypothesis, a the candidate action, and $\hat{F}^{k,a}$ the landmark set H_k predicts after executing a . The transition W applies one shared projection routine to every hypothesis and differs only in the world each asserts: a panel paints content on the screen plane, so that content gains no differential parallax; a mirror preserves the parallax of virtual structure while adding moving virtual images of observer-attached markers; a slab displaces content along the optical axis. The projection is shared rather than implemented per hypothesis so that any difference between predictions is attributable to the mechanism and not to how each was rendered.

3.3 SEPARABILITY, IDENTIFIABILITY, AND ABSTENTION

Pairwise separability is the distance between the consequences two hypotheses predict for the same action,

$$\Delta_{ij}(a) = D(\hat{F}^{i,a}, \hat{F}^{j,a}), \quad (3)$$

where i and j index two hypotheses and D is a composite distance,

$$D = \lambda_m D_{\text{motion}} + \lambda_o D_{\text{occlusion}} + \lambda_g D_{\text{geometry}} + \lambda_f D_{\text{feature}}. \quad (4)$$

In equation 4, D_{motion} is the root-mean-square image-space displacement between the two predicted landmark sets over jointly visible landmarks, in pixels; $D_{\text{occlusion}}$ is the number of landmarks whose predicted visibility differs; D_{geometry} is the root-mean-square three-dimensional distance between back-projected landmarks, in metres; D_{feature} is reserved for a learned feature distance; and $\lambda_m, \lambda_o, \lambda_g, \lambda_f$ weight these terms. Throughout, $\lambda_m = 1$ and $\lambda_g = \lambda_f = 0$, placing D in pixels so the threshold below is a perceptual displacement threshold in the same units; the geometry term is dropped as largely redundant with the motion term once the projection is shared. The occlusion term counts rather than takes a fraction, since a fraction would be diluted by the landmark count and whether a decisive cue cleared the threshold would then depend on how densely a scene was sampled; its weight is set so one unambiguous appearance or disappearance clears it, frequently the only cue a mirror offers.

The quantity of interest is the best separability the action set can achieve,

$$\mathcal{I}_{\mathcal{A}}(H_i, H_j) = \max_{a \in \mathcal{A}} \Delta_{ij}(a), \quad (5)$$

so a pair counts as separable if any single permitted motion would distinguish it. The pipeline these pieces form is drawn in Figure 6. The system abstains when

$$\min_{(i,j)} \mathcal{I}_{\mathcal{A}}(H_i, H_j) < \epsilon, \quad (6)$$

the minimum running over pairs still in contention, with ϵ the perceptual threshold. The order matters: a maximum over actions nested inside a minimum over pairs licenses an answer only if some action separates every undecided pair, and separating most is insufficient because the pair left unseparated is exactly where an error would be made. The threshold is fixed at $\epsilon = 1.0$ px and is not tuned, being a property of the sensor and the perceptual criterion; altering it changes what “identifiable” denotes and relabels the benchmark rather than improving performance on it. The headline real-data measure is rank-based and so invariant to it.

3.4 CHOOSING THE INTERVENTION

Two objectives are compared. The first is the summed objective standard in active perception,

$$a_{\text{sum}}^* = \arg \max_{a \in \mathcal{A}} \sum_{i < j} p_i p_j \Delta_{ij}(a), \quad (7)$$

where p_i and p_j are posterior probabilities, so still-plausible pairs contribute more than decided ones. Such a sum can be maximised by an action separating pairs not in contention while leaving the deciding pair unseparated, since a large term compensates for a zero one. The second objective replaces the sum with the weakest contending pair,

$$a_{\text{maximin}}^* = \arg \max_{a \in \mathcal{A}} \min_{(i,j) \in \mathcal{C}} w_{ij} \Delta_{ij}(a), \quad (8)$$

where $\mathcal{C}$ collects pairs whose joint posterior mass has not collapsed and w_{ij} weights a pair by that mass; decided pairs are excluded rather than down-weighted so the objective tracks only what remains undecided. The form follows from the abstention rule, since a decision requires every contending pair to be separated and the binding quantity is therefore the smallest separability rather than the sum. This criterion is T -optimality from optimal design for model discrimination (Atkinson & Fedorov, 1975; Hunter & Reiner, 1965), applied here to active visual perception.

Which quantity is available, and when. The two objectives above are stated over hypothesis pairs, so they require at least two pairs to differ from one another. That holds in the synthetic benchmark, where four mechanisms give six pairs and the weakest-pair form is therefore the operative criterion. A calibrated stereo pair supports two mechanisms and hence a single pair, so on real photographs equation 7 and equation 8 coincide and neither is informative about the other. Selection on real data is instead driven by the belief update of equation 10: among the viewpoints an observer could occupy, the one to prefer is the one at which some hypothesis best accounts for the view actually obtained,

$$a_{\text{evidence}}^* = \arg \min_{a \in \mathcal{A}} \min_k e_k(a), \quad (9)$$

where $e_k(a)$ is the residual of hypothesis k after executing a , as defined in Section 3.5. Both rules act on the same belief machinery, one through the separability that enters it and the other through the residual that leaves it, and each is used where the evidence it needs exists: equation 8 where several pairs contend, equation 9 where the observation itself must arbitrate. Section 4.4 reports the second on real photographs.

3.5 BELIEF UPDATE, OUTPUT, AND REQUIRED BASELINE

After the selected action is executed and O' observed, each hypothesis has residual $e_k = D(\hat{F}^{k,a^*}, E(O'))$ under the same distance equation 4, so prediction error and separability share a scale. Belief is updated multiplicatively,

$$p_{t+1}(H_k) \propto \exp(-\beta e_k) p_t(H_k), \quad (10)$$

where $p_t(H_k)$ is the prior mass on hypothesis k , e_k its residual in pixels, and β converts error into log-likelihood; $\beta = 1.0$ makes a one-pixel error cost one nat, tying the update to the same perceptual scale as ϵ . A floor on each posterior prevents one unlucky observation from permanently eliminating a hypothesis that later evidence would support. The system then names the mechanism and reports its contact geometry, the depth of the first physical surface along each ray, which is the glass for a mirror and the panel for a display, since that is the quantity determining the cost of an error. Commitment is withheld when the posterior maximum does not exceed $\tau = 0.45$, set just below the 0.5 tie a forced choice over a two-way ambiguity produces so that such a choice counts as false certainty rather than caution; a sweep over $\tau \in \{0.40, \dots, 0.90\}$ is recorded so the value is never load-bearing.

3.6 BENCHMARK AND METRICS

Each base scene is instantiated under every mechanism with content adjusted so the reference views coincide, and the maximum reference-view deviation is of order 10^{-14} px, which is numerical precision. Chance-level single-view performance is therefore a verified property of the benchmark rather than an assumption carried into the results. Configurations that no permitted action resolves, namely view-tracked displays observed within their aperture and mirrors whose geometry mimics an opening, are included deliberately: without them abstention cannot be scored, and a method that never abstains would be indistinguishable from one that abstains correctly. Splits are formed by base scene rather than by variant, so variants of one scene never span partitions.

Four quantities are reported. Causal explanation accuracy (CEA) is $P(\hat{H} = H^*)$ with abstention counted as an error, and committed accuracy is $P(\hat{H} = H^* \mid \text{answered})$; the first penalises excessive caution and the second rewards it, so together they expose the trade between answering and being right. False physical certainty (FPCR) is $P(\max_k p(H_k) > \tau \mid y_{\text{id}} = 0)$, measured only on cases no permitted action resolves, and is what abstention exists to reduce. Normalised regret compares the chosen action's utility against the best available.

3.7 PROTOCOL FOR REAL PHOTOGRAPHS

A calibrated rectified stereo pair supplies exactly one executed action with known parameters, the baseline translation between the cameras, so real data is the case $|\mathcal{A}| = 1$ and equation 5 reduces to the separability of that action. Five elements prevent the measurement from absorbing the effect under study.

Calibration outside, scoring inside. A monocular predictor returns relative inverse depth, defined up to unknown scale s and shift t , which are fitted by least squares against reference disparity on the non-ambiguous region and applied inside it, giving the disparity predicted under H_D ,

$$d_D = s\hat{p} + t, \quad (11)$$

with $\hat{p}$ the raw prediction. Determining the two free parameters where the predictor performs well prevents them from absorbing the discrepancy in the region under study, which is what makes this a test rather than a curve fit.

Both hypotheses come from the predictor's own estimate. The disparity under H_E is the least-squares plane through d_D over the ambiguous region, fitted to the prediction and never to the refer-

ence. Fitting to the reference would place ground truth inside a quantity that must be computable at test time, and since admissibility below selects regions whose reference is already near-planar, that plane would collapse onto the reference and Δ would silently become a measure of predictor error rather than of separability.

Separability is measured photometrically. The right image is warped into the left frame under each hypothesis using the rectified-stereo relation that column x with disparity d corresponds to column $x - d$ in the other view,

$$R_k(x, y) = I_{\text{right}}(x - d_k(x, y), y), \quad k \in \{D, E\}, \quad (12)$$

and separability is the median absolute difference between the two warps inside the region,

$$\Delta_{DE} = \text{median} |R_D - R_E|. \quad (13)$$

Both terms of equation 13 warp the same photograph rather than comparing against the opposite view, so illumination, sensor noise and rectification error appear in both and cancel. Intensity units are deliberate: a disparity disagreement is visible only where the image carries structure to reveal it, so on a featureless region two hypotheses predict different geometry and identical pixels, and the scene is unidentifiable at any baseline.

Ground truth is admitted only when it can arbitrate. A printed sheet or display panel is flat, so reference disparity inside the region should be planar. Where it is not, stereo matching has locked onto displayed content rather than the panel and cannot settle whether the predictor reported the panel or the depicted scene. An image is retained only when a plane fitted to reference disparity inside the region attains $R^2 \geq 0.90$, and the number excluded is reported, being itself a finding about the evaluation data.

Failure is defined without reference to any signal under test. A predictor is recorded as fooled when its relative error inside the region exceeds its error outside, a label mentioning neither confidence nor identifiability, so neither can be fitted to it. Three quantities requiring no model and no theory are reported alongside Δ_{DE} : the fraction of the frame the region occupies, the local horizontal image gradient, and the raw disagreement $|d_D - d_E|$ before photometric weighting. A proposed measure unable to outperform these has not earned its complexity, and reporting them is what makes that visible. All AUROC values use the convention that a larger score indicates greater likelihood of error, so a value below 0.5 appears as anti-correlation rather than being concealed by inverting the score.

A second dataset supplies ordinal relations between point pairs rather than dense depth, in two parallel sets: one ordering by the depth of the first surface along the ray, one ignoring transparency. Pairs appearing in both with opposite order are the discriminating cases, since only there do the two readings predict oppositely. Its per-point layer index, authored independently of this work, serves as an external label of ambiguity sharing no machinery with the measures above.

3.8 LEARNED ABSTENTION AND MODELS EVALUATED

Two policies are evaluated, both leaving the depth predictor frozen and fitting only a small model on top, since the predictor’s behaviour is the object of study and updating it would change what is measured. The first uses the hand-designed quantities above as features. The second pools the network’s own frozen intermediate features over the ambiguous region, keeping the difference between the pooled representation inside and outside it together with the within-region spread; the contrast is used because a representation encoding ambiguity should differ inside the region relative to its surroundings, whereas an absolute vector would largely encode which scene is viewed.

Three families of classifier are fitted so that the result does not rest on one inductive bias: L_2 -regularised logistic regression on standardised features, histogram gradient boosting, and XG-BOOST. The boosted models are constrained to depth 3 with a minimum of 40 samples per leaf, because a few thousand rows over fewer than a hundred scenes will otherwise be memorised, and the logistic head applied to the pooled encoder features is strongly regularised because its feature vector is far wider than the sample count. Reporting all three also makes the comparison auditable: agreement between families indicates that the measured quantity is a property of the data rather than of the optimiser.

Three protocol choices make these interpretable. Folds are split by base scene, because the benchmarks photograph one scene from several viewpoints and a random split would place near-duplicates on both sides and report memorisation as transfer. Leave-one-model-out trains on all checkpoints but one and tests on the held-out one, asking whether failure is predictable as a property of the image rather than of one network. Hyperparameter selection is nested, with settings chosen on inner folds and scored only on outer folds unseen during selection, since a single-level sweep reporting its own best score would be optimistically biased by more than the margin claimed. The region-occupancy baseline is retained as an input feature rather than removed, so a policy relying chiefly on it is revealed by permutation importance instead of flattered by its exclusion.

Published checkpoints are used as released, spanning several architecture families and roughly two orders of magnitude in parameter count, with nothing trained or tuned on any evaluation set. One implementation detail affects correctness: the output attribute conventionally named `predicted_depth` carries metric depth in some families and inverse depth in others, so each checkpoint’s convention is verified against reference stereo disparity before use, an inverted convention exchanging near for far while still producing plausible numbers.

4 RESULTS

4.1 SYNTHETIC BENCHMARK

Table 1 reports Phase 1 over 11 seeds with the benchmark redrawn per seed, so intervals reflect the data-generating process rather than model-side noise. The single-view classifier sits at exactly chance in every seed, confirming that the matched construction leaves nothing for a single view to exploit, and the ordering $\text{single-view} < \text{passive} < \text{intervention-aware}$ holds strictly in 11/11 seeds individually. Abstention is what removes false certainty: the same system without it answers confidently on unresolvable scenes at 0.832 and with it at 0.000, while committed accuracy rises from 0.722 to 0.855.

Table 2 isolates the choice of intervention. In both studies the summed objective equation 7 falls short on mirrors while maximin equation 8 reaches 1.000, because a belief-weighted sum can be maximised by an action that separates pairs not in contention and leaves the deciding pair unseparated. The gap is structural rather than a threshold choice, since an action scoring zero on a pair cannot separate it at any ϵ .

4.2 REAL PHOTOGRAPHS

Twelve published checkpoints spanning six architecture families and a $38\times$ range in parameter count are evaluated as released (Figure 3a,b; per-checkpoint values in Table 5). Every one is fooled on 59.5–78.4% of admissible images, the newest family among the most affected, and on every one the model’s own confidence scores below chance at predicting its own failure (0.234–0.465), while identifiability ranges 0.467–0.719 and exceeds confidence on all twelve.

Calibration cannot repair a signal ordered against the quantity of interest. Both signals meet their split-conformal coverage guarantee at all five levels tested, yet deferring on confidence leaves a retained set failing at 76.6% against a 65.2% base rate, while deferring on identifiability lowers it to 54.4% (Figure 3d, Table 4).

Two further measurements support the protocol. The admissibility criterion of Section 3.7 excludes 159 of 455 images whose stereo reference locks onto displayed content rather than the panel; on admissible images where a monitor fills the frame, error inside the region is $2.22\times$ that outside and every image is fooled, with reference planarity 0.989 against 0.821 for the prediction. Independently, on a second dataset with ordinal rather than dense supervision, accuracy split by the benchmark authors’ own per-point layer index gives 0.848 against 0.737 across 306,874 pairs, and a second architecture shows a larger gap still (0.843 against 0.669; Table 10). That label shares no machinery with any measure defined here.

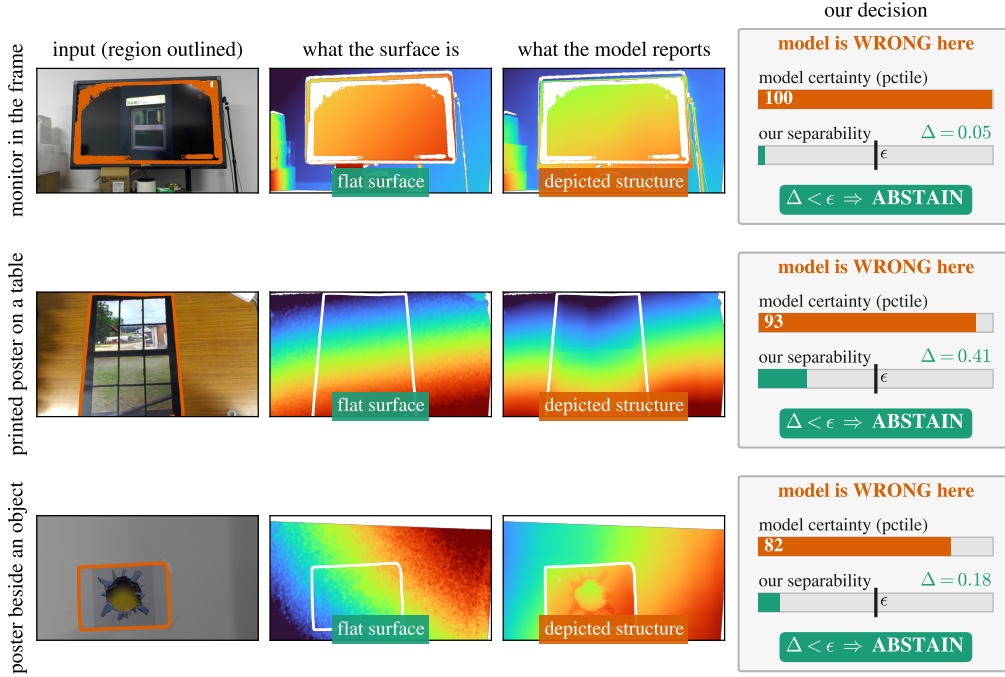


Figure 2: **Why the decision is needed.** Each row is an image the model gets wrong: the reference shows a flat sheet or panel, the model reports the structure depicted on it. Being wrong costs it no certainty, and in the top row it is the most confident prediction of all 296 images. The separability Δ available from the executed motion falls below ϵ in every case, so equation 6 declines to answer. Across the benchmark 82 of 296 images are failures where the model ranks in the most-confident 30% and this rule abstains.

Table 1: **Phase 1, 11 seeds, mean $\pm$ std, chance 0.333.** CEA scores abstention as an error, so committed accuracy is the like-for-like column. Removing hypothesis conditioning collapses every separability to zero and the system abstains everywhere, the correct failure mode for a load-bearing component.

Method	CEA	Committed	Abst.	AUROC	FPCR
Single-view classifier	0.333 $\pm$ 0.000	0.333 $\pm$ 0.000	0%	0.505 $\pm$ 0.066	0.000 $\pm$ 0.000
Passive multi-view	0.621 $\pm$ 0.046	0.621 $\pm$ 0.046	0%	0.748 $\pm$ 0.065	0.868 $\pm$ 0.107
Generic NBV proxy	0.371 $\pm$ 0.050	0.706 $\pm$ 0.029	48%	0.898 $\pm$ 0.026	0.000 $\pm$ 0.000
Random intervention	0.409 $\pm$ 0.056	0.730 $\pm$ 0.027	44%	0.925 $\pm$ 0.024	0.000 $\pm$ 0.000
Max-baseline intervention	0.502 $\pm$ 0.061	0.824 $\pm$ 0.019	39%	0.963 $\pm$ 0.028	0.000 $\pm$ 0.000
<i>abl.</i> no hypothesis cond.	0.000 $\pm$ 0.000	—	100%	0.500 $\pm$ 0.000	0.000 $\pm$ 0.000
<i>abl.</i> noisy encoder	0.557 $\pm$ 0.034	0.848 $\pm$ 0.049	34%	0.994 $\pm$ 0.016	0.008 $\pm$ 0.026
<i>abl.</i> no abstention	0.722$\pm$0.029	0.722 $\pm$ 0.029	0%	1.000 $\pm$ 0.000	0.832 $\pm$ 0.107
Ours	0.560 $\pm$ 0.032	0.855$\pm$0.054	34%	1.000 $\pm$ 0.000	0.000$\pm$0.000

4.3 CHOOSING WHERE TO LOOK, ON REAL PHOTOGRAPHS

The benchmark photographs each static scene from several camera positions, and the median stereo disparity differs across those frames while the scene itself does not, which is what a moved observer produces. Fifty-five scenes therefore offer a real action set of three or more viewpoints, and selection can be evaluated on photographs rather than in simulation. Applying equation 9, and measuring the relative error inside the ambiguous region so that the outcome depends on no other part of the image, choosing the best-explained viewpoint reduces that error by **31.6%** relative to choosing uniformly at random, and does so on **12** of 12 checkpoints; the proportion of images on which the predictor is fooled falls from 48.7% to 44.1%, on 9 of 12. Table 11 gives the full comparison. What is

Table 2: **Choosing the intervention, 11 seeds per study.** Under forced choice maximin exceeds the passive baseline by +0.297 ($t = 41.6$, 11/11 seeds). Chance is 0.250 in Phase 2.

Objective	Mirror committed		Phase 2, four mechanisms		
	selector study	Phase 2	committed	abstains	forced CEA
Single-view classifier	—	—	0.250±0.000	0%	0.250±0.000
Passive multi-view	—	—	0.510±0.021	0%	0.510±0.021
Max-baseline	—	0.887	0.930±0.019	73%	—
Summed equation 7	0.587±0.172	0.956±0.037	0.986±0.011	69%	—
Maximin equation 8	1.000±0.000	1.000±0.000	1.000±0.000	72%	0.808±0.024

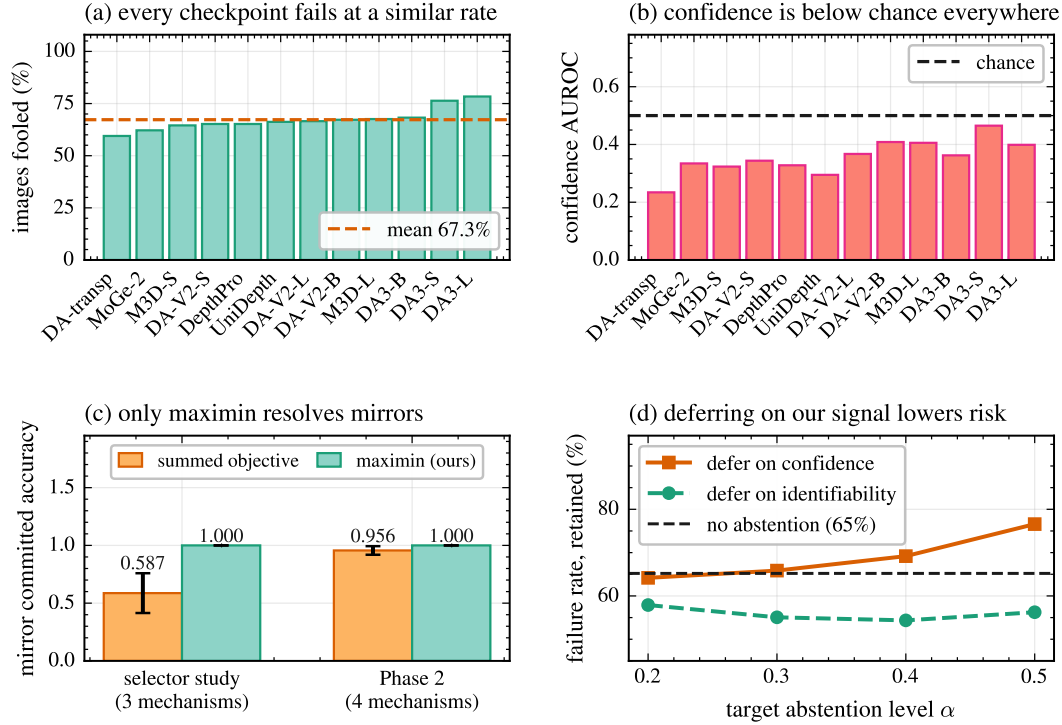


Figure 3: **Headline results.** (a) The proportion of images each checkpoint is fooled on stays within a narrow band despite a $38\times$ range in parameter count, so the behaviour is structural rather than a capacity limitation. (b) Every checkpoint’s own confidence lies below chance at predicting its own failure. (c) Mirror committed accuracy in both selector studies; only maximin reaches 1.000. (d) Split-conformal selective risk: both signals meet coverage, but only deferring on identifiability lowers the failure rate.

measured is the error at the chosen viewpoint rather than the recovery of the mechanism itself, since real images carry no mechanism annotation; it is the closest test the data supports.

4.4 LEARNED ABSTENTION

The signals above are hand-designed, so a natural question is whether a learned combination does better and whether what it learns is specific to one predictor. Under leave-one-model-out, a gradient-boosted policy trained on eleven checkpoints and tested on the twelfth reaches **0.880** against 0.764 for the strongest single feature, ahead on **12** of 12 held-out checkpoints (Figure 4a). Nested selection, choosing hyperparameters on inner folds and scoring only on outer folds never seen during selection, moves the result by under 0.01, and XGBOOST and histogram boosting agree within 0.002, so the figures are not artefacts of tuning. On a single checkpoint, where every method sees identical data, a head reading *frozen* encoder features reaches **0.879** against 0.858 for ten hand-designed

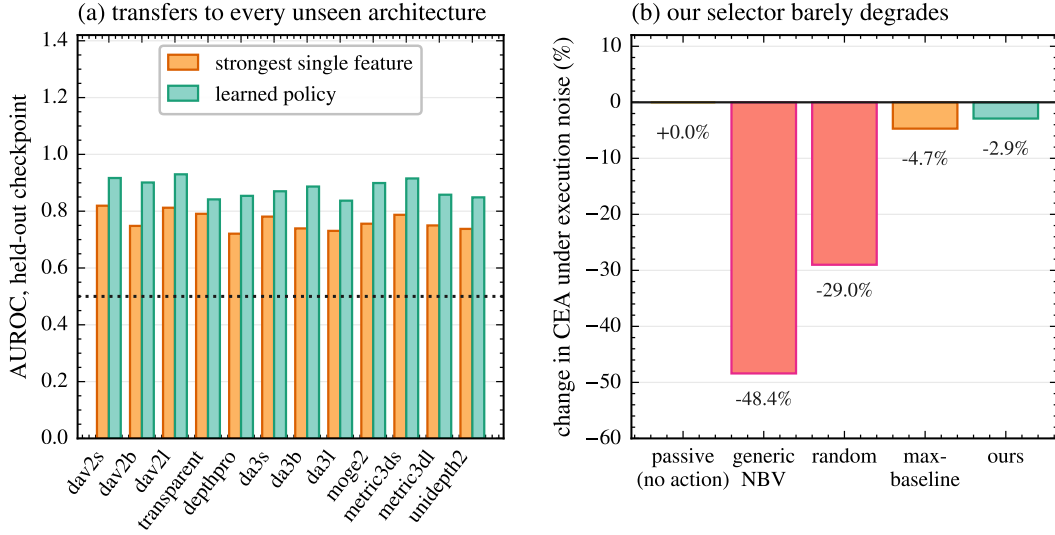


Figure 4: **Transfer and stability.** (a) A policy trained on every checkpoint but one and evaluated on the checkpoint it has never seen, against the strongest single feature on identical folds; the learned policy is ahead on all twelve. (b) Change in accuracy when the chosen action is executed imperfectly.

quantities and 0.819 for the strongest single feature: evidence of ambiguity is already present in the representation and available to be decoded. Table 7 gives the full comparison.

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SUPPLEMENTARY MATERIAL

Every number reported here is drawn from the same runs as the main paper. Nothing in this appendix revises a result in Section 4; the tables below extend those results with the remaining metrics, the per-family breakdown, and the analytical component omitted from the main text for space.

A WHY ONE IMAGE ADMITS SEVERAL MECHANISMS

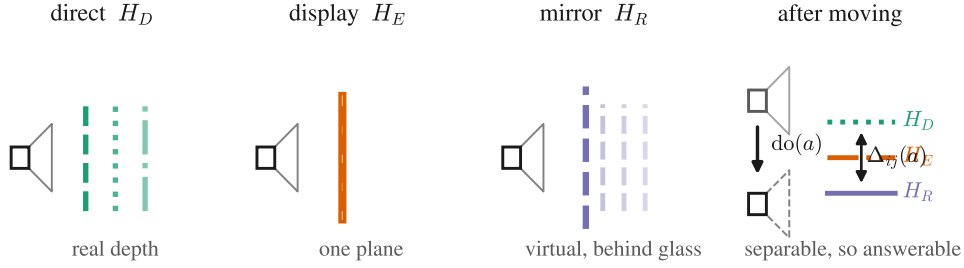


Figure 5: **One image, three mechanisms, and what motion reveals.** A real corridor, the same corridor painted on a display, and its virtual image in a mirror can be arranged to produce identical pixels at the reference view, so no single-view method separates them. They differ in the surface a hand would touch, shown as the coloured plane in each panel. Executing an action makes their predictions diverge, and the size of that divergence is the separability $\Delta_{ij}(a)$ of equation 3.

B THE DECISION PIPELINE

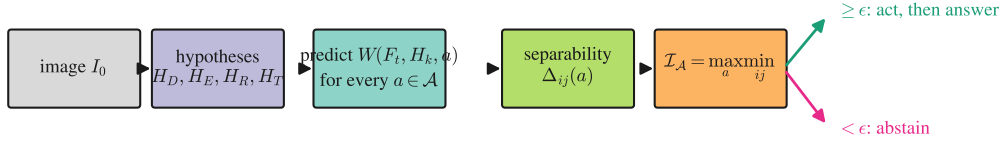


Figure 6: **How the pieces compose.** Each hypothesis predicts the consequence of every permitted action equation 2; pairwise separabilities equation 3 are formed from those predictions; identifiability is the best value the action set attains on the weakest contending pair equation 5. Only when it clears the perceptual threshold is an action executed and an answer given, and otherwise the system declines equation 6. The whole comparison is made before any motion, which is what allows abstention in advance of acting.

C THE REAL-PHOTOGRAPH PROTOCOL IN DETAIL

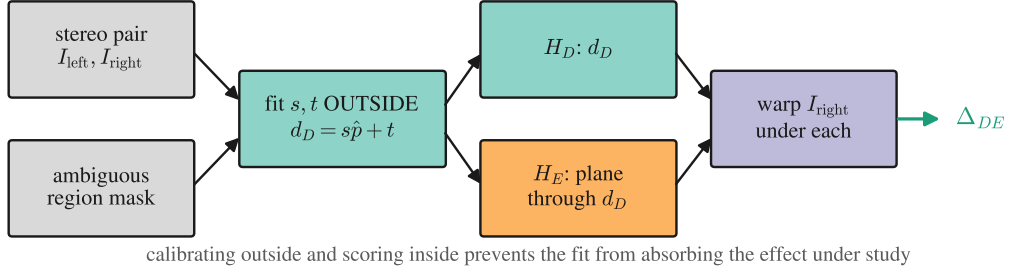


Figure 7: **Construction of the real-data measurement.** Scale and shift are fitted on the non-ambiguous region and applied inside it, so the two free parameters are determined where the predictor performs well and cannot absorb the discrepancy in the region under study. Both hypotheses are then generated from the predictor’s own aligned disparity, never from the reference measurement, and the right image is warped into the left frame under each. Comparing the two warps rather than either against the opposite view cancels illumination, sensor noise and rectification error, which appear identically in both.

D MINIMUM CAUSAL RESOLVING BASELINE

For the pair consisting of a direct scene and a static planar display, the motion required to separate them has a closed form. Under a lateral translation of baseline B , a point at depth Z shifts in the image by approximately fB/Z , where f is the horizontal focal length in pixels, so two points at depths $Z_1 \neq Z_2$ produce a differential parallax of $fB|1/Z_1 - 1/Z_2|$. A static planar display produces none once the screen-plane homography is compensated. Requiring the differential parallax to exceed the perceptual threshold δ gives

$$B_{\min} > \frac{\delta}{f|1/Z_1 - 1/Z_2|}, \quad (14)$$

where δ coincides with ϵ whenever the composite distance is dominated by its motion term. Equation equation 14 is valid only for pure lateral translation without rotation, a pinhole camera, small baselines satisfying $B \ll Z$, and this hypothesis pair alone. It does not apply to view-tracked displays, which no baseline resolves, nor to mirrors, whose virtual scene carries genuine parallax. These conditions are checked before the expression is evaluated, so a violated assumption yields a missing prediction rather than a confident wrong one.

E CALIBRATED DEFERRAL AND IMPERFECT EXECUTION

Two studies are reported here: split-conformal deferral on the real benchmark, and the effect of executing a chosen action imperfectly on the synthetic one.

Table 3: **Executing the chosen action imperfectly, paired over 11 seeds (± 1 cm, $\pm 0.5^\circ$).** Methods that take no action are unchanged at exactly 0.0%, the check that perturbation cannot reach a method that does not act. Among methods that do act, the hypothesis-blind proxy loses roughly half its accuracy while ours loses 2.9%: actions chosen with a large separability margin tolerate perturbation, and that margin is what equation 8 maximises.

Method	Clean CEA	Perturbed CEA	Change	t
Passive multi-view (does not act)	0.621	0.621	+0.0%	—
Generic NBV proxy	0.371	0.192	−48.4%	−14.05
Random intervention	0.409	0.291	−29.0%	−10.64
Max-baseline	0.502	0.478	−4.7%	−3.68
Ours	0.560	0.544	−2.9%	−3.05

Table 4: **Split-conformal selective prediction, 296 images, 400 calibration splits per level.** Both signals meet their coverage guarantee at every level, so neither procedure is broken and the separation appears entirely in selective risk, which coverage does not constrain. Deferring on confidence tracks the base rate and then worsens, reaching 76.6%; deferring on identifiability lowers the failure rate at every level, to 54.4% at its best.

Deferral signal	AUROC	Failure rate among retained images				Coverage	Best gain
		$\alpha=0.2$	$\alpha=0.3$	$\alpha=0.4$	$\alpha=0.5$		
No abstention	—	65.2%				—	—
Confidence (TTA)	0.339	64.2%	65.8%	69.2%	76.6%	5/5	−1.0
Identifiability	0.632	57.9%	55.1%	54.4%	56.2%	5/5	−10.8

F PER-CHECKPOINT RESULTS

Table 5 gives the per-checkpoint values summarised in Section 4.2, with parameter counts and family membership.

Table 5: **Twelve published checkpoints, six families, used as released.** Every checkpoint is fooled on 59.5–78.4% of admissible images, and the range narrows neither with parameter count nor with recency. On every checkpoint the model’s own confidence scores *below chance* at predicting its own failure, so the usual deferral rule is inverted with respect to what it should track. Confidence is the better of raw confidence and a four-member test-time-augmentation ensemble.

Checkpoint	Family	Params	Foiled	Confidence AUROC	Ident. AUROC
DA-V2-Small	Depth-Anything-V2	24.8M	65.2%	0.344	0.645
DA-V2-Base	Depth-Anything-V2	97.5M	67.2%	0.408	0.588
DA-V2-Large	Depth-Anything-V2	335.3M	66.6%	0.367	0.467
DA-transparency	Depth-Anything-V2	24.8M	59.5%	0.234	0.719
DA3-Small	Depth-Anything-3	24.8M	76.4%	0.465	0.479
DA3-Base	Depth-Anything-3	97.5M	68.2%	0.362	0.576
DA3-Large	Depth-Anything-3	410.9M	78.4%	0.399	0.512
DepthPro	DepthPro	952.0M	65.2%	0.328	0.659
MoGe-2	MoGe-2	330.9M	62.2%	0.334	0.642
Metric3D-Small	Metric3D	37.5M	64.5%	0.323	0.593
Metric3D-Large	Metric3D	343.0M	67.6%	0.406	0.593
UniDepth-v2	UniDepth	353.8M	66.2%	0.295	0.692
<i>range</i>	<i>6 families</i>	<i>38×</i>	<i>59.5–78.4%</i>	<i>0.234–0.465</i>	<i>0.467–0.719</i>

G EXTENDED RESULTS BY ARCHITECTURE FAMILY

Averaging within architecture family confirms that the pattern is not driven by one lineage.

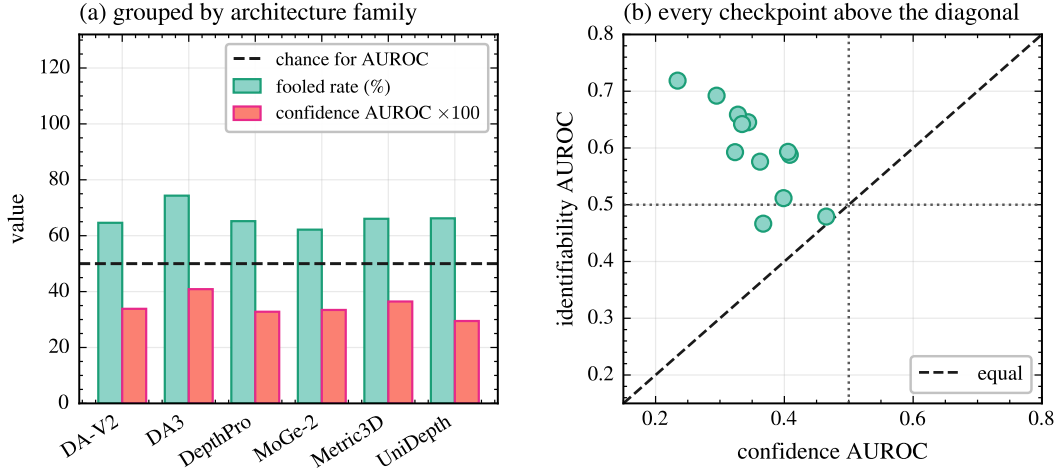


Figure 8: **Grouped and paired views of Table 5.** (a) Averaged within each architecture family, the fooled rate sits well above the confidence AUROC in every family, and every family’s confidence average falls below chance. (b) The same twelve checkpoints plotted against one another: each point lies above the diagonal, so identifiability exceeds confidence on every checkpoint without exception, and every point lies to the left of the vertical chance line.

Table 6: **Per-family summary of the twelve checkpoints**, averaged within family. The pattern of Table 5 is not driven by one lineage: every family is fooled on roughly two images in three, and every family’s mean confidence AUROC lies below chance.

Family	Checkpoints	Mean fooled	Mean confidence	Mean identifiability
Depth-Anything-V2	4	64.6%	0.338	0.605
Depth-Anything-3	3	74.3%	0.409	0.522
DepthPro	1	65.2%	0.328	0.659
MoGe-2	1	62.2%	0.334	0.642
Metric3D	2	66.1%	0.365	0.593
UniDepth	1	66.2%	0.295	0.692
All	12	67.3%	0.355	0.597

H LEARNED ABSTENTION, SUMMARY

The table below is the summary referred to from Section 4.4; the section after it expands the leave-one-model-out column to every held-out checkpoint.

Table 7: **Learned abstention.** Folds are split by base scene, so frames of one scene never span a partition. Success under leave-one-model-out means a property of the image was learned rather than of one predictor.

Policy	Grouped CV	Nested CV	Leave-one-model-out
<i>all 12 checkpoints, 3,552 image-model pairs, 68 scenes</i>			
Strongest single feature	0.762	—	0.764
Logistic regression	0.787	0.748	0.808
XGBOOST	—	0.832	—
Histogram gradient boosting	0.832	0.834	0.880 (12/12)
<i>single checkpoint, identical data for all three</i>			
Strongest single feature	0.819	—	—
Hand-designed features	0.858	—	—
Frozen-encoder head	0.879	—	—

I FULL LEAVE-ONE-MODEL-OUT RESULTS

The leave-one-model-out protocol of Table 7 is expanded to every held-out checkpoint below, together with the regularisation sweep for the frozen-encoder head.

Table 8: **Learned abstention in full.** *Upper block:* every held-out checkpoint from Table 7. The policy is trained on the other eleven and evaluated on the one named, which it has never seen; it exceeds the strongest single feature on all twelve, with a mean margin of +0.115 and a minimum of +0.051. *Lower block:* the frozen-encoder head against regularisation strength. It varies by 0.026 across three decades and exceeds the hand-designed feature set (0.858) at every setting, so the result reflects the representation rather than the choice of constant.

Held out	Learned	Baseline	Δ	Held out	Learned	Baseline	Δ
DA-V2-Small	0.917	0.819	+0.098	DA3-Large	0.837	0.731	+0.106
DA-V2-Base	0.901	0.748	+0.153	DepthPro	0.854	0.721	+0.133
DA-V2-Large	0.930	0.812	+0.118	MoGe-2	0.899	0.756	+0.143
DA-transparency	0.842	0.791	+0.051	Metric3D-Small	0.915	0.787	+0.128
DA3-Small	0.870	0.781	+0.089	Metric3D-Large	0.858	0.750	+0.108
DA3-Base	0.887	0.739	+0.147	UniDepth-v2	0.849	0.738	+0.111
mean 0.880 against 0.764, margin +0.115, ahead on 12/12							
<i>frozen-encoder head: AUROC against regularisation strength C, grouped five-fold, one checkpoint</i>							
C	0.001	0.005	0.020	0.100	0.500	2.000	10.000
AUROC	0.869	0.878	0.879	0.872	0.864	0.859	0.853

J PER-CATEGORY BEHAVIOUR ON THE ILLUSION BENCHMARK

The aggregate error ratio quoted in Section 4.2 resolves into three categories that behave quite differently.

Table 9: **Admissible images by illusion category.** The penalty for reporting depicted rather than physical geometry is largest where a display fills the frame, and the reference measurement inside these regions is close to perfectly planar while the prediction is not, confirming that the surface is flat and the prediction is not. This is the per-category detail behind the aggregate quoted in Section 4.2.

Category	Images	Error ratio inside/outside	Foiled	Reference planarity
Printed sheet on a table	97	0.78×	33%	0.99
Poster beside an object	104	1.47×	63%	0.99
Monitor filling the frame	95	2.23×	100%	0.99
All admissible	296	1.42×	65.2%	0.989

K SECOND DATASET WITH ORDINAL SUPERVISION

The second dataset supplies ordinal relations between point pairs rather than dense depth, and carries a per-point layer index authored independently of this work.

Table 10: **LayeredDepth validation, 299 images, 306,874 annotated point pairs.** Accuracy is split by the benchmark authors’ own per-point layer index, a label written independently of this work that shares no machinery with any measure defined here. Both architectures are markedly worse on points visible only through a transparent surface, and the penalty is larger for the model with more parameters. The ambiguity this paper formalises is therefore recognised by an independent annotation and independently predicts failure, on supervision that is ordinal rather than dense and on a dataset collected by a different group.

Checkpoint	Single-layer acc.	Multi-layer acc.	Gap	Ordinal acc. (first surface)	Ordinal acc. (through glass)
DA-V2-Small	0.848	0.737	+0.111	0.847	0.784
DepthPro	0.843	0.669	+0.175	0.842	0.741

L CHOOSING WHERE TO LOOK

Fifty-five non-video scenes offer three or more camera positions of the same static content. Video scenes are excluded because their frames differ by what is displayed rather than by where the observer stands, so treating them as an action set would count a change of stimulus as a change of viewpoint.

Table 11: **Viewpoint selection on real photographs**, 12 checkpoints over 55 scenes, lower is better on both columns. Selecting by equation 9, the viewpoint at which some hypothesis best accounts for the observed view, lowers the error inside the ambiguous region by 31.6% against a random choice and wins on every checkpoint. Error inside the region is the primary column because it depends on no other part of the image; the fooled rate is reported alongside for continuity with the rest of the paper.

Viewpoint chosen by	Error inside region	Fooled
first frame, no choice	0.0695	49.2%
uniformly at random	0.0610	48.7%
the predictor’s own confidence	0.0450	47.1%
best-explained observation equation 9	0.0417	44.1%
<i>ahead of a random choice on 12/12 checkpoints by error; 9/12 by fooled rate</i>		

A note on the comparison with published methods. The benchmark’s authors release both their data and a trained checkpoint. The datasets carry an explicit Apache-2.0 licence and are used here accordingly; the checkpoint carries no licence field on its model repository, no LICENSE file on either default branch of the source repository, and no licence statement in its README, all checked on 2026-08-29. A direct comparison against it is therefore deferred until those terms are stated, rather than made under unclear ones.

M METRIC DEFINITIONS

The metrics used throughout are defined below.

Table 12: **The reported metrics.** $\hat{H}$ is the predicted mechanism, H^* the true one, $p(H_k)$ the posterior mass on hypothesis k , τ the confidence threshold of Section 3.5, and $y_{\text{id}} = 0$ marks a case no permitted action resolves. CEA and committed accuracy are reported together because the first penalises excessive caution and the second rewards it.

Metric	Definition	Question it answers
CEA	$P(\hat{H} = H^*)$	how often the mechanism is named correctly, abstention counted as an error
Committed acc.	$P(\hat{H} = H^* \mid \text{answered})$	when an answer is given, how often it is right
Abstention rate	$P(\text{abstain})$	how often an answer is withheld
FPCR	$P(\max_k p(H_k) > \tau \mid y_{\text{id}} = 0)$	how often a confident answer is given where none is warranted
Ident. AUROC	predicted against true resolvability	whether unresolvable cases are recognised
Normalised regret	chosen against best available action	whether a near-optimal intervention was selected
Motion cost	translation plus weighted rotation	how much movement the decision required

N SCOPE AND LIMITATIONS

Four limits bound what the reported numbers establish, and each is visible in the results rather than inferred.

The synthetic identifiability AUROC is degenerate by construction. On the synthetic benchmark the computation producing the identifiability prediction is the same one producing the ground-truth label, so the value of 1.000 in Table 1 carries no information about the method’s accuracy. It is reported only as a contrast against the classifiers, which reach 0.505 and 0.748 on identical scenes, and no claim rests on its absolute value.

Abstention is not unconditionally safe under encoder noise. The clean configuration reaches a false physical certainty rate of 0.000 ± 0.000 across all 11 seeds, but the noisy-encoder ablation reaches 0.008 ± 0.026 : one seed in eleven produces a false certainty. The correct statement is therefore “zero in 10 of 11 seeds under encoder noise”, not “zero”.

Real-data evidence rests on stereo, which supplies a single action. A calibrated pair gives $|\mathcal{A}| = 1$, so equation 5 reduces to one term and the selection results of Table 2 cannot be tested on this data. Evaluating the choice of intervention on real photographs requires a multi-view or controllable-camera capture, which the datasets used here do not provide.

One checkpoint runs in a degraded configuration. UniDepth-v2 has no optimised attention kernel for the available hardware and falls back to a slower path, so its numbers in Table 5 are a lower bound on the released model rather than a tuned comparison. It is retained because excluding a checkpoint on the basis of its result would bias the sweep.

O REPRODUCIBILITY

All synthetic results are averaged over 11 seeds with the benchmark redrawn per seed, so the reported intervals reflect the data-generating process rather than model-side noise alone. Splits are formed by base scene throughout, in both the synthetic benchmark and the learned-policy protocols, so variants or frames of a single scene never span a partition. Every depth checkpoint is used exactly as released and nothing is trained or tuned on any evaluation set. Each checkpoint’s output convention is verified against reference stereo disparity before use, since the attribute conventionally named `predicted_depth` carries metric depth in some families and inverse depth in others. One checkpoint, UniDepth-v2, runs without its optimised attention kernel on the available hardware, so its numbers should be read as a lower bound on the released model rather than a tuned comparison.